

UNIT-III

Dielectric, Magnetic and Energy of Materials

Dielectrics materials

Introduction: Dielectrics are insulating materials so there are no free charge carriers in them.

Ex : Mica, plastic glass and wood.

The dielectrics are of two types they are,

1)Non-polar dielectrics :

In case of non-polar dielectrics the centre of gravity of positive charge is coincide with the center of gravity of negative charge.

They have symmetric in nature.

Their dipole moment is zero.

Ex: methane, benzene etc.

2)Polar-dielectrics :

In case of polar dielectrics the center of gravity of positive charge is not coincide with the center of gravity of negative charge.

They have asymmetric in nature.

Their dipole moment is not zero.

Ex: NH₃, Hcl, water etc.

3.1 Basic definitions

Electric dipole: A system consisting of two equal and opposite charge separated by a small distance is called as electric dipole.

Length of dipole is denoted by $2l$ (or) d .

$q \text{ O} \text{-----} \text{O} \text{ -}q$

$\leftarrow 2l \rightarrow$

Electric dipole moment: The product of magnitude of charge and the distance between two charges is called as electric dipole moment It is denoted by P (or) μ

Dipole moment(μ) = charge $\times$ distance between them

$$(\mu) = q \times 2l$$

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Units are coulomb-metre

Dielectric constant(or) relative permittivity (ϵ_r): Ratio between the permittivity of the medium to the permittivity of free space is called as relative permittivity.

That is (ϵ_r) = permittivity of medium / permittivity of free space

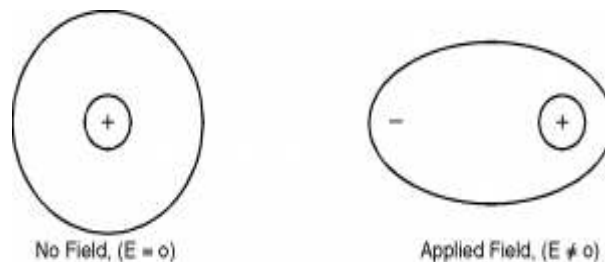
$$(\epsilon_r) = \epsilon / \epsilon_o = \text{capacitance of capacitor with dielectric} / \text{capacitance of capacitor without dielectric}$$

$$= c / c_o$$

It has no units

Electric polarization

The process of producing electric dipoles which are oriented along the field direction is called as polarization in dielectrics. It is denoted by P



Polarizability(α):

The induced dipole moment is directly proportional to the intensity of the electric field.

That is μ proportional E

$$\mu = \alpha E \text{ where } \alpha \text{ is called proportionality constant of polarizability}$$

Induced dipole moment per unit electric field is called as polarizability

$$\text{i.e., } \alpha = \mu / E$$

Polarization vector ($\vec{P}$): The dipole moment per unit volume of the dielectric material is called as polarization vector P.

$$\vec{P} = N \mu$$

Where μ = average dipole moment per molecule

N = no. of molecules per unit volume

Electric flux density (or) Electric displacement (D):

Electric displacement at a point in a material is given by $D = \epsilon E$

We know that $\epsilon = \epsilon_o \epsilon_r$

$$D = \epsilon_o \epsilon_r E$$

As the polarization measures the additional flux density rising from the presence of the material as compared to free space is given by

$$D = \epsilon_o E + P$$

We know that $D = \epsilon_o \epsilon_r E$

$$\epsilon_o \epsilon_r E = \epsilon_o E + P$$

$$P = \epsilon_o \epsilon_r E - \epsilon_o E$$

$$P = \epsilon_o (\epsilon_r - 1) E$$

Electric susceptibility(χ_e):

The polarization vector is proportional to the total electric flux density(E)

$$\text{i.e., } P = \epsilon_o \chi_e E$$

$$\chi_e = P / \epsilon_o E$$

$$\chi_e = \epsilon_o (\epsilon_r - 1) E / \epsilon_o E$$

$$\chi_e = (\epsilon_r - 1)$$

3.2 Types of polarization

They are four types of polarizations

- 1) Electronic polarization
- 2) Ionic polarization
- 3) Orientation polarization and
- 4) Space charge polarization

3.2.1 Electronic polarization:

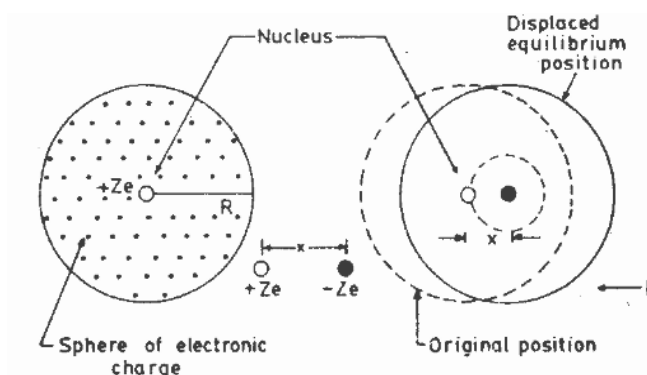
The displacement of the positively charged nucleus and the electrons of an atom in opposite directions, on application of an electric field, result in electronic polarization.

Induced dipole moment μ proportional E

$$\mu = \alpha_e E$$

where α_e is the electronic polarizability. It is independent of temperature.

Calculation of electronic polarizability:



A simplified classical model of an atom is shown in above figure. Here the nucleus of charge Ze is surrounded by an electron cloud of charge $-Ze$ distributed in a sphere of radius R . The charge density ρ is given by

$$\begin{aligned} \rho &= \frac{-Ze}{\frac{4\pi R^3}{3}} \\ &= -3Ze / 4\pi R^3 \end{aligned}$$

When an external field of intensity E is applied, the nucleus and the electrons experience Lorentz forces of magnitude ZeE in opposite directions. Hence the nucleus and electron cloud are pulled apart. When they are separated a coulomb force developed between them, which tends to oppose the displacement. When these forces namely Lorentz force and coulomb force are equal and opposite, equilibrium is reached and let x be the displacement under that condition.

$$\text{Lorentz force} = -ZeE$$

$$\text{Coulomb force} = Ze(\text{charge enclosed in the sphere of radius } x) / 4\pi\epsilon_0 x^2$$

$$\text{The charge enclosed in the sphere} = \frac{4\pi x^3}{3} \rho$$

$$= (4\pi x^3 / 3) (- 3Ze / 4\pi R^3)$$

$$= - Zex^3 / R^3$$

$$\text{Hence coulomb force} = (Ze / 4\pi\epsilon_0 x^2) (-Zex^3 / R^3) = - Z^2 e^2 x / 4\pi\epsilon_0 R^3$$

In the equilibrium position, Lorentz force = coulomb force

$$- ZeE = - Z^2 e^2 x / 4\pi\epsilon_0 R^3$$

$$X = 4\pi\epsilon_0 R^3 E / Ze$$

The two electric charges Ze and $-Ze$ are separated by a distance x under the action of the applied field thus constituting induced electric dipoles.

Induced electric dipole moment $\mu_e = Zex$

$$= Ze4\pi\epsilon_0 R^3 E / Ze$$

$$= 4\pi\epsilon_0 R^3 E$$

$$\text{i.e., } \mu_e = \alpha_e E$$

Where $\alpha_e = 4\pi\epsilon_0 R^3$ is called electronic polarizability.

The dipole moment per unit volume is called electronic polarization. It is independent of temperature.

$\mathbf{P} = N \mu_e = N \alpha_e E$ where N is Number of atoms / m^3

$$\mathbf{P}_e = N (4 \pi \epsilon_0 R^3 E) = 4 \pi \epsilon_0 R^3 N E \text{ where}$$

R is radius of atom

3.2.2 Ionic Polarization

It is due to the displacement of cat ions and anions in opposite directions and occurs in an ionic solid .

Consider a NaCl molecule. Suppose an electric field is applied in the positive direction . The positive ion moves by x_1 and the negative ion moves by x_2

Let M is mass of positive ion

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m is mass of negative ion

Total displacement $x = x_1 + x_2$ ----- (1)

Lorentz force on positive ion $= +eE$ -----(2) Lorentz force on negative ion $= -eE$

----- (3) Restoring force on positive ion $= -k_1 x_1$ ----- (2 a)

Restoring force on negative ion $= +k_2 x_2$ ----- (3 a) where k_1, k_2 Restoring force constants

At equilibrium, Lorentz force and restoring force are equal and opposite

For positive ion, $eE = k_1 x_1$

$$X_1 = Ee / k_1$$

For negative ion, $eE = k_2 x_2$

$$X_2 = Ee / k_2$$

Where $k_1 = M \omega_0^2$ & $k_2 = m \omega_0^2$ where ω_0 is angular velocity of ions

Therefore $x = x_1 + x_2 = (eE / \omega_0^2) [1/M + 1/m]$ ----- (4)

From definition of dipole moment

$\mu = \text{charge} \times \text{distance of separation}$

$$\mu = e x = (e^2 E / \omega_0^2) [1/M + 1/m]$$

But $\mu \propto E$ or $\mu = \alpha_i E$

Therefore $\alpha_i = (e^2 / \omega_0^2) [1/M + 1/m]$ --- (5)

This is ionic polarization

3.2.3 Orientational Polarization:

In methane molecule, the centre of negative and positive charges coincides, so that there is no permanent dipole moment. On the other hand, in certain molecules such as CH_3Cl , the positive and negative charges do not coincide. Even in the absence of an electric field, this molecule carries a dipole moment, they tend to align themselves in the direction of applied field. The polarization due to such alignment is called orientation polarization. It is dependent on temperature. With increase of temperature the thermal energy tends to randomize the alignment.

$$\begin{aligned}\text{Orientational polarization } P_o &= N\mu = N\mu^2 E / 3KT \\ &= N \alpha_o E\end{aligned}$$

Therefore Orientational polarizability $\alpha_o = P_o / NE$

$$= \mu^2 / 3kT$$

Thus orientational polarizability α_o is proportional to absolute temperature material.

3.2.4 Space-charge polarization:

Space-charge polarization occurs due to the accumulation of charge at the electrodes or at the interfaces in a multiphase material.

The total polarization of a material is the sum of the contribution from the various sources seen above.

$$P_{\text{total}} = P_e + P_i + P_o + P_s$$

Since the space-charge polarizability is very small when compared to other types of polarizabilities, the total polarizability of a gas can be written as

$$\begin{aligned}\alpha &= \alpha_e + \alpha_i + \alpha_o \\ &= 4\pi\epsilon_o R^3 + (e^2 / \omega_o^2) [1/M + 1/m] + \mu^2 / 3kT\end{aligned}$$

Hence total polarization is given by

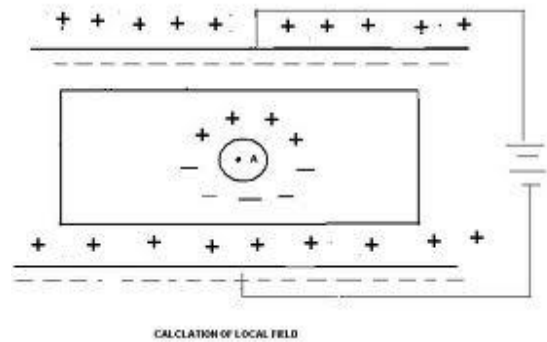
$$P = N\alpha E$$

$$P = NE \{ 4\pi\epsilon_o R^3 + (e^2 / \omega_o^2) [1/M + 1/m] + \mu^2 / 3kT \}$$

This equation is known as Langevin-Debye equation

3.3 Internal fields in solids: (Lorentz method):

Let a dielectric be placed between the plates of a parallel plates capacitor and let there be an imaginary spherical cavity around the atom A inside the dielectric. It is also assumed that the radius of the cavity is large compared to the radius of the atom.



The internal fields at the atom site A can be considered to be made up of the following four components namely E_1 , E_2 , E_3 , and E_4 .

$$E_{\text{int}} = E_1 + E_2 + E_3 + E_4 \quad \text{-----(1)}$$

E_1 is the field intensity at A due to the charge density on the plates. From the field theory

$$E_1 = D / \epsilon_0 \quad \text{we know that } D = P + \epsilon_0 E$$

$$E_1 = P + \epsilon_0 E / \epsilon_0 = E + \frac{P}{\epsilon_0} \quad \text{----- (2)}$$

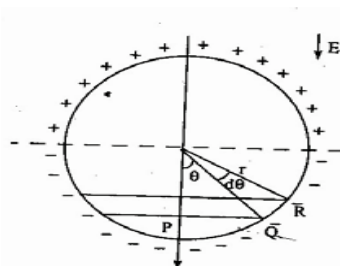
E_2 is the field intensity at A due to the charge density induced on the two sides of the dielectric.

$$\text{Therefore } E_2 = - P / \epsilon_0 \quad \text{----- (3)}$$

E_3 is the field intensity at A due to other atoms contained in the cavity. We are assuming a cubic structure, because of symmetry

$$E_3 = 0 \quad \text{----- (4)}$$

Calculation of E_4 :



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If dA is the surface area of the sphere of radius r lying between θ and $\theta + d\theta$, where θ is the direction with reference to the direction of the applied force,

$$\text{then } dA = 2\pi(PQ)(QR)$$

$$\text{but } \sin\theta = PQ / r, PQ = r \sin\theta$$

$$\text{and } d\theta = QR / r, QR = r d\theta$$

$$\text{Hence } dA = 2\pi r \sin\theta r d\theta = 2\pi r^2 \sin\theta d\theta$$

The charge dq on the surface dA is equal to the normal component of the polarization multiplied by the surface area.

$$\text{Therefore } dq = P \cos\theta dA = P (2\pi r^2 \sin\theta \cos\theta d\theta)$$

The field due to this charge at A , denoted by dE_4 in the direction $\theta = 0$ is

$$\begin{aligned} dE_4 &= dq \cos\theta / 4\pi\epsilon_0 r^2 \\ &= P \cos\theta 2\pi r^2 \sin\theta \cos\theta d\theta / 4\pi\epsilon_0 r^2 \\ &= P \cos^2\theta \sin\theta d\theta / 2\epsilon_0 \end{aligned}$$

Thus the total field E_4 due to the charges on the surface of the entire cavity is obtained by integrating

$$\begin{aligned} E_4 &= \int dE_4 = \frac{P}{2\epsilon_0} \int_0^\pi \cos^2\theta \sin\theta d\theta \\ \text{let } \cos\theta &= x \\ \sin\theta d\theta &= -dx \end{aligned}$$

limits are, when $\theta = 0^\circ$, $x = 1$ and $\theta = 180^\circ$, $x = -1$

$$\begin{aligned} E_4 &= \frac{P}{2\epsilon_0} \int_{-1}^1 x^2 (-dx) \\ E_4 &= \frac{P}{2\epsilon_0} \int_{-1}^1 x^2 dx \\ E_4 &= \frac{P}{2\epsilon_0} \left[\frac{x^3}{3} \right]_{-1}^1 \\ E_4 &= \frac{P}{2\epsilon_0} \left[\frac{1}{3} - \left(-\frac{1}{3}\right) \right] \\ E_4 &= \frac{P}{2\epsilon_0} \left(\frac{2}{3} \right) \\ E_4 &= \frac{P}{3\epsilon_0} \text{ ----- (5)} \end{aligned}$$

Substituting the equations (2),(3),(4) and (5) in equation (1)

$$\text{We get } E_{\text{int}} = E + \frac{P}{\epsilon_0} - \frac{P}{\epsilon_0} + \frac{P}{3\epsilon_0}$$

$$E_{\text{int}} = E + \frac{P}{3\epsilon_0} \quad \text{this is the equation for internal field}$$

3.4 Clausius – Mosotti Relation:

Let us consider the elemental dielectric having cubic structure. Since there are no ions and permanent dipoles in these materials, then ionic polarizability α_i and orientational polarizability α_0 are zero.

$$\text{i.e. } \alpha_i = \alpha_0 = 0$$

$$\text{Hence polarization } P = N \alpha_e E$$

$$= N \alpha_e (E + P / 3\epsilon_0)$$

$$\text{i.e. } P [1 - N \alpha_e / 3 \epsilon_0] = N \alpha_e E$$

$$P = N \alpha_e E / [1 - N \alpha_e / 3 \epsilon_0] \text{ ---(1)}$$

We know that

$$D = P + \epsilon_0 E$$

$$P = D - \epsilon_0 E$$

Dividing on both sides by E

$$P / E = D / E - \epsilon_0 = \epsilon - \epsilon_0 = \epsilon_0 \epsilon_r - \epsilon_0$$

$$P = E \epsilon_0 (\epsilon_r - 1) \text{ -----(2)}$$

From eqn1 and 2, we get

$$P = E \epsilon_0 (\epsilon_r - 1) = N \alpha_e E / [1 - N \alpha_e / 3 \epsilon_0]$$

$$[1 - N \alpha_e / 3 \epsilon_0] = N \alpha_e / [\epsilon_0 (\epsilon_r - 1)]$$

$$1 = N \alpha_e / 3 \epsilon_0 + N \alpha_e / \epsilon_0 (\epsilon_r - 1)$$

$$1 = (N \alpha_e / 3 \epsilon_0) (1 + 3 / (\epsilon_r - 1))$$

$$1 = (N \alpha_e / 3 \epsilon_0) [(\epsilon_r - 1) + 3] / (\epsilon_r - 1)$$

$$1 = (N \alpha_e / 3 \epsilon_0) [(\epsilon_r + 2) / (\epsilon_r - 1)]$$

$$(\epsilon_r + 2) / (\epsilon_r - 1) = N \alpha_e / 3 \epsilon_0$$

Where N – no of molecules per unit volume

This is Clausius – Mosotti Relation.

3.5 Piezo – Electricity:

These materials have the property of becoming electrically polarized when mechanical stress is applied. This property is known as Piezo – electric effect has an inverse

.According to inverse piezo electric effect, when an electric stress or voltage is applied, the material becomes strained. The strain is directly proportional to the applied field E.

When piezo electric crystals are subjected to compression or tension, opposite kinds of charges are developed at the opposite faces perpendicular to the direction of applied force. The charges produced are proportional to the applied force.

Piezo – Electric Materials and Their Applications: Single crystal of quartz is used for filter, resonator and delay line applications. Natural quartz is now being replaced by synthetic material.

Rochelle salt is used as transducer in gramophone pickups, ear phones,

hearing aids, microphones etc. the commercial ceramic materials are based on barium titanate, lead zirconate and lead titanate. They are used for high voltage generation (gas lighters),

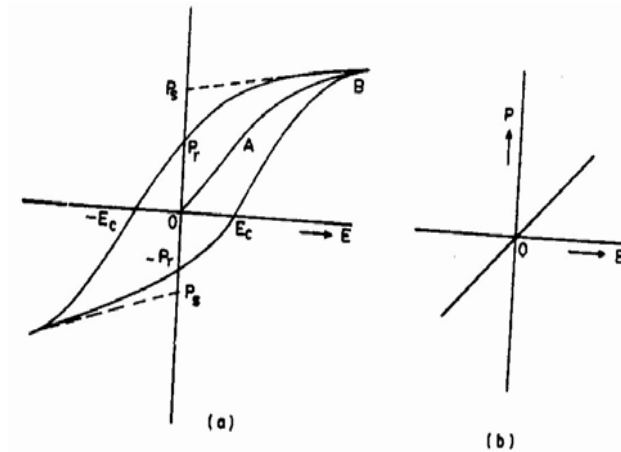
accelerometers, transducers etc.

3.6 Ferro electricity:

Ferro electric materials are an important group not only because of intrinsic Ferro electric property, but because many possess useful piezo electric, birefringent and electro optical properties.

The intrinsic Ferro electric property is the possibility of reversal or change of orientation of the polarization direction by an electric field. This leads to hysteresis in the polarization P, electric field E relation , similar to magnetic hysteresis. Above a critical

temperature, the Curie point T_c , the spontaneous polarization is destroyed by thermal disorder. The permittivity shows a characteristic peak at T_c .



3.7 PYRO ELECTRICITY

Pyroelectric effect is the change in spontaneous polarization when the temp. of the specimen is changed. The pyro electric co-efficient λ is defined as the change in polarization per unit temperature change of the specimen.

$$\lambda = \frac{dp}{dT}$$

Change in polarization results in change in external field and hence charge on the surface. As the possible to detect a change of 16×10^{-16} with a suitable electrometer temp. changes as small as $6 \times 10^{-6} \text{ c}$ – can be measured using the pyro electric effect.

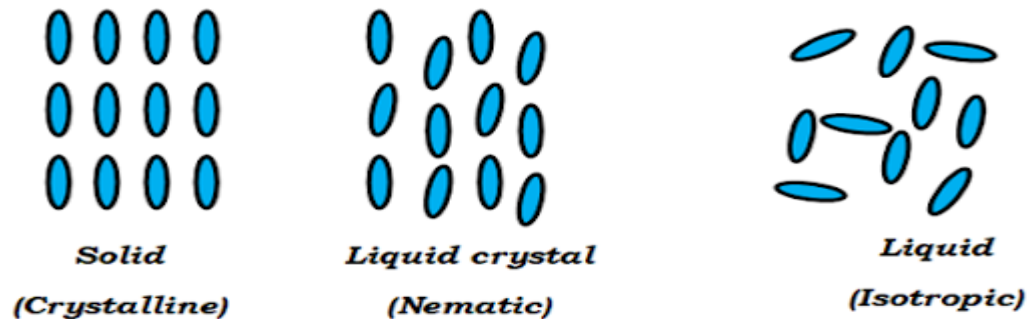
Applications

The pyro electric materials such as BaTiO_3 , LiNbO_3 etc are used to make very good instead detectors which can operator at room temp. materials such as TGS, NaNO_2 and PZT ceramics are used in polarizing and construction of pyro electric image tubes.

3.8 Liquid Crystal Display

Liquid Crystal Display (LCD) is a flat display screen used in electronic devices such as laptop, computer, TV, cellphones and portable video games. As the name says liquid crystal is a material which flows like a liquid and shows some properties of solid. These LCD are very thin displays and it consumes less power than LEDs.

Molecular arrangement of Liquid Crystal:

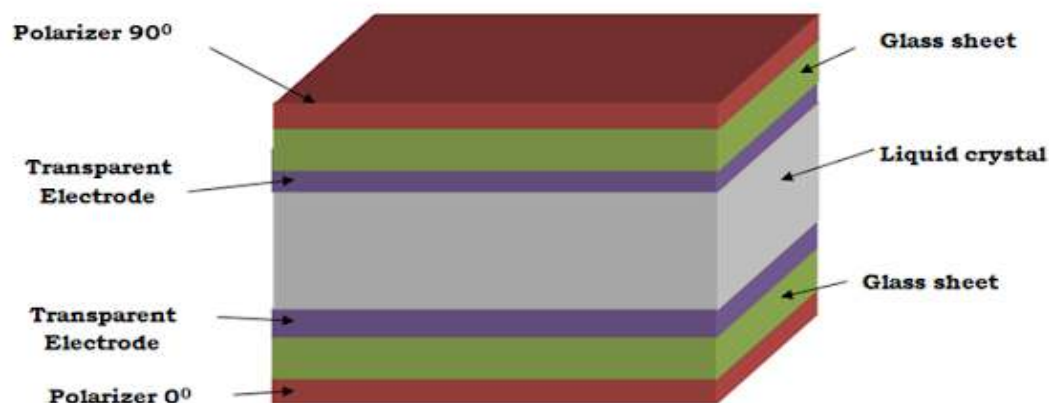


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Molecular arrangement of liquid crystal

As the name says the molecular structure of liquid crystal is in between solid crystal and liquid isotropic. In Liquid crystal display (LCD) nematic type of liquid crystal molecular arrangement is used in which molecules are oriented in some degree of alignment. For example when we increase the temperature the ice cube melts and liquid crystal is like the state in between ice cube and water.

Construction of Liquid Crystal Display:



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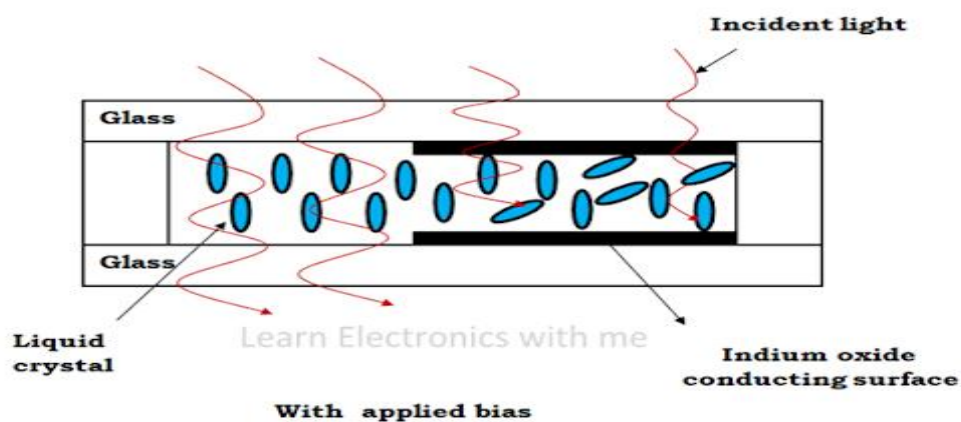
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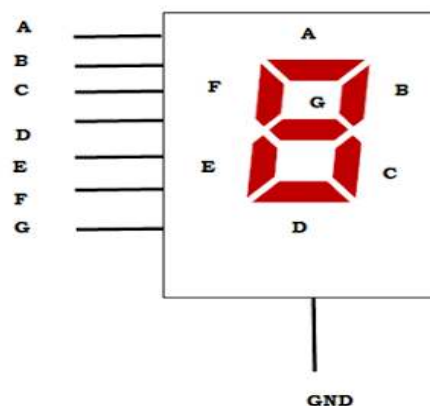
Construction of LCD consists of two polarized glass pieces. Two electrodes are used, one is positive and the other one is negative. External potential is applied to LCD through this electrodes and it is made up of indium-tin-oxide. Liquid crystal layer of about $10\mu\text{m}$ - $20\mu\text{m}$ is placed between two glass sheets. The light is passed or blocked by changing the polarization.

Working of Liquid Crystal Display

The basic working principle of LCD is blocking of light. It does not produce light on its own. So external light source is used. When the external light passes from one polarizer to the next polarizer, external supply is given to the liquid crystal, the polarized light aligns itself so that the image is produced in the screen.



The indium oxide conducting surface is a transparent layer which is placed on both the sides of the sealed thick layer of liquid crystal. When no external bias is applied the molecular arrangement is not disturbed.



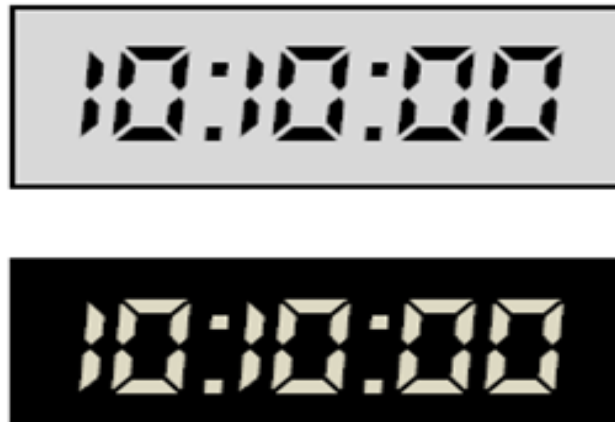
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When the external bias is applied the molecular arrangement is disturbed and it and that area looks dark and the other area looks clear.

In the segment arrangement, the conducting segment looks dark and the other segment looks clear. To display number 2, the segments A,B,G,E,D are energized.

Positive and Negative LCDs:



In positive LCD display the segments are dark and the background is white and the polarizers are placed perpendicular to each other. In the negative LCD display the segments are white in the dark background and the polarizers are aligned to each other.

Advantages:

- It is thin and compact
- Low power consumption
- Less heat is emitted during operation
- Low cost

Disadvantages:

- Speed of operation is low
- Lifespan is less
- Restricted viewing angles

Applications:

- Used in digital wrist watch
- Display images in digital cameras
- Used in numerical counters
- Display screen in calculators
- Mainly used in television
- Used in mobile screens

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- Used in video players
- Used in image sensing circuits

Magnetic materials

The basic aim in the study of the subject of magnetic materials is to understand the effect of an external magnetic field on a bulk material, and also to account for its specific behaviour. A dipole is an object that a magnetic pole is on one end and an equal and opposite second magnetic dipole is on the other end.

A bar magnet can be considered as a dipole with a north pole at one end and South Pole at the other. If a magnet is cut into two, two magnets or dipoles are created out of one. This sectioning and creation of dipoles can continue to the atomic level. Therefore, the source of magnetism lies in the basic building block of all the matter i.e. the atom.

Consider electric current flowing through a conductor. When the electrons are flowing through the conductor, a magnetic field is formed around the conductor. A magnetic field is produced whenever an electric charge is in motion. The strength of the field is called the **magnetic moment**.

Magnetic materials are those which can be easily magnetized as they have permanent magnetic moment in the presence of applied magnetic field. Magnetism arises from the magnetic dipole moments. It is responsible for producing magnetic influence of attraction or repulsion.

3.9 Basic definitions

Magnetic dipole : it is a system consisting of two equal and opposite magnetic poles separated by a small distance of '2l' metre.

Magnetic Moment (μ_m) : It is defined as the product of the pole strength (m) and the distance between the two poles (2l) of the magnet.

$$\text{i.e.} \dots \mu_m = (2l) m$$

Units: Ampere – metre²

Magnetic Flux Density or Magnetic Induction (B): It is defined as the number of magnetic lines of force passing perpendicularly through unit area.

$$\text{i.e.} \dots B = \text{magnetic flux} / \text{area} = \Phi / A \text{ Units: Weber} / \text{metre}^2 \text{ or Tesla.}$$

Permeability:

Magnetic Field Intensity (H): The magnetic field intensity at any point in the magnetic field is the force experienced by a unit north pole placed at that point.

Units: Ampere / meter

The magnetic induction B due to magnetic field intensity H applied in vacuum is related by

B proportional

$$B = \mu_0 H$$

where μ_0 is permeability of free space = $4 \pi \times 10^{-7} \text{ H / m}$

If the field is applied in a medium, the magnetic induction in the solid is given by

$$B = \mu H$$

where μ is permeability of the material in the medium

$$\mu = B / H$$

Hence magnetic Permeability μ of any material is the ratio of the magnetic induction to the applied magnetic field intensity. The ratio of μ / μ_0 is called the relative permeability (μ_r).

$$\mu_r = \mu / \mu_0$$

$$\text{Therefore } B = \mu_0 \mu_r H$$

Magnetization: It is the process of converting a non – magnetic material into a magnetic material. The intensity of magnetization (M) of a material is the magnetic moment per unit volume.

intensity of magnetization (M) = magnetic moment / volume

The intensity of magnetization is directly related to the applied field H through the susceptibility of the medium (χ) by

$$\chi = M / H \text{ ----- (1)}$$

The magnetic susceptibility of a material is the ratio of the intensity of magnetization produced to the magnetic field intensity which produces the magnetization. It has no units.

We know

$$B = \mu H$$

$$= \mu_0 \mu_r H$$

$$\text{i.e } B = \mu_0 \mu_r H + \mu_0 H - \mu_0 H$$

$$= \mu_0 H + \mu_0 H (\mu_r - 1)$$

$$= \mu_0 H + \mu_0 M \quad \text{where } M \text{ is magnetization} = H (\mu_r - 1)$$

$$\text{i.e } \mathbf{B} = \mu_0 (\mathbf{H} + \mathbf{M}) \text{ -----(2)}$$

The first term on the right side of eqn (2) is due to external field. The second term is due to the magnetization.

$$\text{Hence } \mu_0 = \mathbf{B} / \mathbf{H} + \mathbf{M}$$

Relative Permeability ,

$$\mu_r = \mu / \mu_0 = (\mathbf{B} / \mathbf{H}) / (\mathbf{B} / \mathbf{H} + \mathbf{M}) = \mathbf{H} + \mathbf{M} / \mathbf{H} = 1 + \mathbf{M} / \mathbf{H}$$

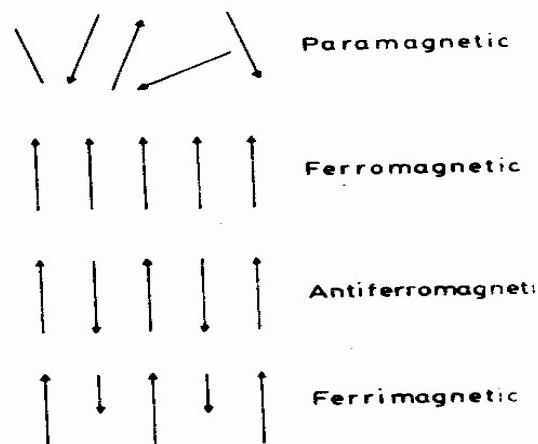
$$\mu_r = 1 + \chi \text{ -----(3)}$$

The magnetic properties of all substances are associated with the orbital and spin motions of the electrons in their atoms. Due to this motion, the electrons become elementary magnets of the substance. In few materials these elementary magnets are able to strengthen the applied magnetic field , while in few others , they orient themselves such that the applied magnetic field is weakened.

3.10 Classification Of Magnetic Materials:

All matter respond in one way or the other when subjected to the influence of a magnetic field. The response could be strong or weak, but there is none with zero response ie, there is no matter which is non magnetic in the absolute sense. Depending upon the magnitude and sign of response to the applied field , and also on the basis of effect of temperature on the magnetic properties, all materials are classified broadly under 5 categories.

1. Diamagnetic materials
2. Paramagnetic materials,
3. Ferromagnetic materials
4. Antiferromagnetic materials and
5. Ferrimagnetic materials.



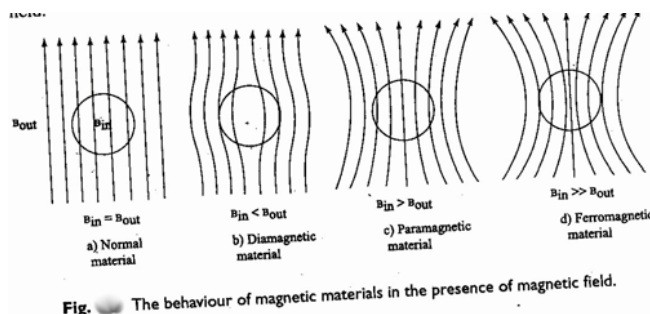
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1. Diamagnetic materials: *Diamagnetic* materials are those which experience a repelling force when brought near the pole of a strong magnet. In a non uniform magnetic field they are repelled away from stronger parts of the field.

In the absence of an external magnetic field, the net magnetic dipole moment over each atom or molecule of a diamagnetic material is zero.

Ex: Cu, Bi, Pb, Zn and rare gases.



Paramagnetic materials: Paramagnetic materials are those which experience a feeble attractive force when brought near the pole of a magnet. They are attracted towards the stronger parts of magnetic field. Due to the spin and orbital motion of the electrons, the atoms of paramagnetic material possess a net intrinsic permanent moment.

Susceptibility χ is positive and small for these materials. The susceptibility is inversely proportional to the temperature T .

$$\chi \propto 1/T$$

$$\chi = C/T \quad \text{where } C \text{ is Curie's temperature.}$$

Below superconducting transition temperatures, these materials exhibit the Paramagnetism. Examples: Al, Mn, Pt, CuCl_2 .

Ferromagnetic Materials: Ferromagnetic materials are those which experience a very strong attractive force when brought near the pole of a magnet. These materials, apart from getting magnetized parallel to the direction of the applied field, will continue to retain the magnetic property even after the magnetizing field is removed. The atoms of ferromagnetic materials also have a net intrinsic magnetic dipole moment which is due to the spin of the electrons.

Susceptibility is always positive and large and it depends upon temperature.

$$\chi = C / (T - \theta) \quad (\text{only in paramagnetic region i.e., } T > \theta)$$

θ is Curie's temperature.

When the temperature of the material is greater than its Curie temperature then it converts into paramagnetic material.

Examples: Fe, Ni, Co, MnO.

Antiferromagnetic matériels : These are the ferromagnetic materials in which equal no of opposite spins with same magnitude such that the orientation of neighbouring spins is in antiparallel manner are present.

Susceptibility is small and positive and it is inversely proportional to the temperature.

$$\chi = C / (T + \theta)$$

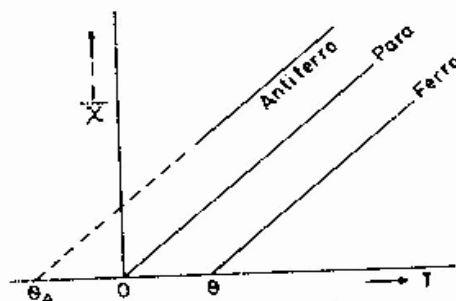
the temperature at which anti ferromagnetic material converts into paramagnetic material is known as Neel's temperature.

Examples: FeO, Cr₂O₃.

Ferrimagnetic materials: These are the ferromagnetic materials in which equal no of opposite spins with different magnitudes such that the orientation of neighbouring spins is in antiparallel manner are present.

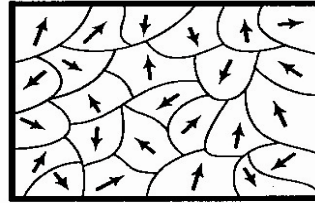
Susceptibility positive and large, it is inversely proportional to temperature

$$\chi = C / (T \pm \theta) \quad T > T_N \text{ (Neel's temperature) Examples : ZnFe}_2\text{O}_4, \text{CuFe}_2\text{O}_4$$

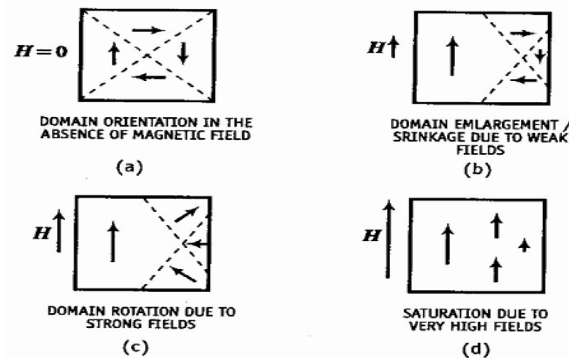


3.11 Domain theory of ferromagnetism:

According to Weiss, a virgin specimen of ferromagnetic material consists of a no of regions or domains ($\approx 10^{-6}$ m or larger) which are spontaneously magnetized. In each domain spontaneous magnetization is due to parallel alignment of all magnetic dipoles. The direction of spontaneous magnetization varies from domain to domain. The resultant magnetization may hence be zero or nearly zero. When an external field is applied there are two possible ways of alignment fo a random domain.



RANDOM ORIENTATION OF MAGNETIC MOMENTS OF THE DOMAINS



i). **By motion of domain walls:** The volume of the domains that are favourably oriented with respect to the magnetizing field increases at the cost of those that are unfavourably oriented

ii) **By rotation of domains:** When the applied magnetic field is strong, rotation of the direction of magnetization occurs in the direction of the field.

3.12 Hysteresis curve (study of B-H curve):

The hysteresis of ferromagnetic materials refers to the lag of magnetization behind the magnetization field. when the temperature of the ferromagnetic substance is less than the ferromagnetic Curie temperature ,the substance exhibits hysteresis. The domain concept is well suited to explain the phenomenon of hysteresis. The increase in the value of the resultant magnetic moment of the specimen by the application of the applied field , it attributes to the 1. motion of the domain walls and 2. rotation of domains.

When a weak magnetic field is applied, the domains that are aligned parallel to the field and in the easy direction of magnetization, grow in size at the expense of less favorably oriented ones. These results in Bloch wall movement and when the weak field is removed, the domains reverse back to their original state. This reverse wall displacement is indicated by OA of the magnetization curve. When the field becomes stronger, the Bloch wall movement continues and

it is mostly irreversible movement. This is indicated by the path AB of the graph. The phenomenon of hysteresis is due to this irreversibility.

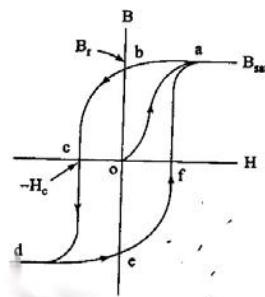
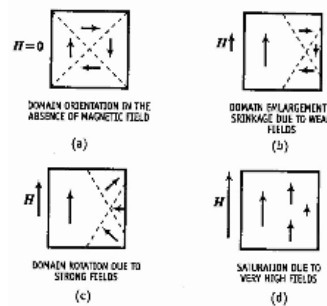


Fig. Schematic representation of the hysteresis loop for a ferromagnetic material.

At the point B all domains have got magnetized along their easy directions. Application of still higher fields rotates the domains into the field direction which may be away from the easy direction. Once the domain rotation is complete the specimen is saturated denoted by C. on removal of the field the specimen tends to attain the original configuration by the movement of Bloch walls. But this movement is hampered by the impurities, lattice imperfections etc, and so more energy must be supplied to overcome the opposing forces. This means that a coercive field is required to reduce the magnetization of the specimen to zero. The amount of energy spent in this regard is a loss. Hysteresis loss is the loss of energy in taking a ferromagnetic body through a complete cycle of magnetization and this loss is represented by the area enclosed by the hysteresis loop.

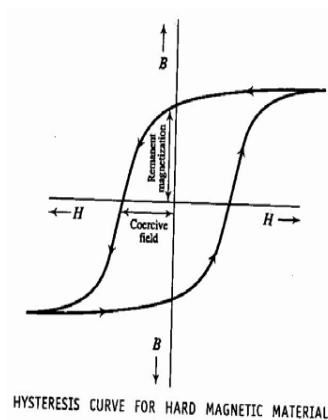
A hysteresis curve shows the relationship between the magnetic flux density B and applied magnetic field H. It is also referred to as the B-H curve(loop).

3.13 Hard and Soft Magnetic Materials:

Hysteresis loop of the ferromagnetic materials vary in size and shape. This variation in hysteresis loops leads to a broad classification of all the magnetic materials into hard type and soft type.

Hard Magnetic Materials:

Hard magnetic materials are those which are characterized by large hysteresis loop because of which they retain a considerable amount of their magnetic energy after the external magnetic field is switched off. These materials are subjected to a magnetic field of increasing intensity, the domain walls movements are impeded due to certain factors. The cause for such a nature is attributed to the presence of impurities or non-magnetic materials, or the lattice imperfections. Such defects attract the domain walls thereby reducing the wall energy. It results in a stable state for the domain walls and gives mechanical hardness to the material which increases the electrical resistivity. The increase in electrical resistivity brings down the eddy current loss if used in a.c conditions. The hard magnetic materials can neither be easily magnetized nor easily demagnetized.

**Properties:**

1. High remanent magnetization
2. High coercivity
3. High saturation flux density
4. Low initial permeability
5. High hysteresis energy loss
6. High permeability
7. The eddy current loss is low for ceramic type and large for metallic type.

Examples of hard magnetic materials are, i) Iron- nickel- aluminum alloys with certain amount of cobalt called Alnico alloy. ii) Copper nickel iron alloys. iii) Platinum cobalt alloy.

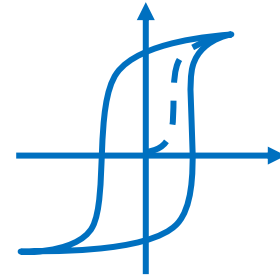
Applications of hard magnetic materials: For production of permanent magnets, used in magnetic detectors, microphones, flux meters, voltage regulators, damping devices and magnetic separators.

SOFT MAGNETIC MATERIALS

- Soft Magnetic materials have low hysteresis loss

due to small hysteresis loop area.

- The curve is tall and thin.
- The coercivity and retentivity are small, hence these materials can be easily magnetized and demagnetized.
- These materials have large values of permeability and susceptibility.
- **Ex: Iron silicon alloys, Ferrous nickel alloy, soft iron etc.,**
- These are used to make electromagnets and mainly used in electro-magnetic machinery and transformer cores and also used in switching circuits , microwave isolators and matrix storage of computers.



3.14 Bubble Memory Devices

It is invented by Andrew Bobeck in 1970s at Bell Labs. Bubble Memory is the storage technology in which thin layer of magnetic material are used to store the information. These magnetic materials will contain small magnetized areas known as bubbles or domains. Each such bubble or domain will store one bit of data. The magnetic material is arranged into parallel tracks so that the bubbles can move along under the application of the external magnetic field. The bubble memory is non-volatile storage technology i.e. the information will retain even after power is switched off.

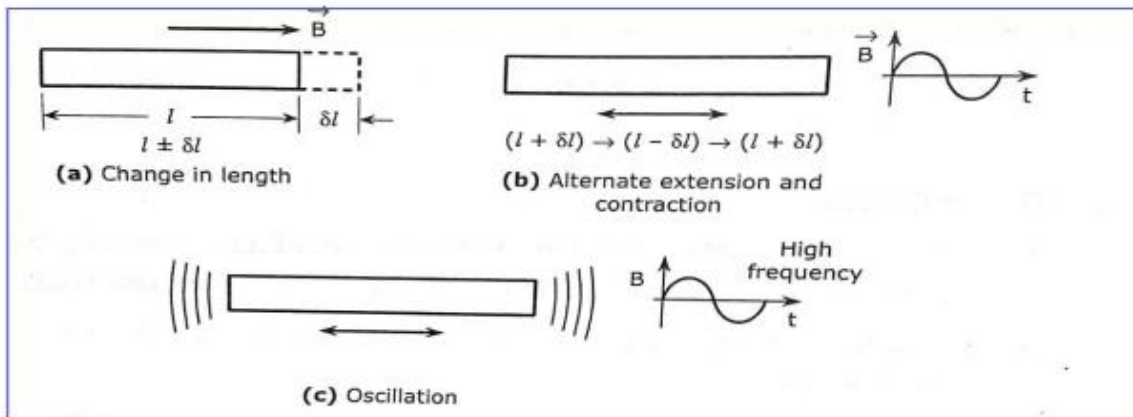
Advantages

1. Low power consumption
2. High functional packing density .
3. More durable than disc memory since it has no moving parts.
4. Rugged in nature which made it important in military applications even though it is not successful in commercial market
5. It is resistant to extreme temperatures, dust, humidity and radiation.

3.15 Magnetostriction (James Joule, 1842)

When a ferromagnetic rod like iron, nickel is kept in the strong magnetic field parallel to its length, it is subjected to a change in dimensions. This is called the magnetostriction effect. If the rod

is placed in a rapidly varying magnetic field alternately expands and contracts with twice the frequency of the applied magnetic field. This change in length of the ferromagnetic material is independent of the polarity of applied magnetic field. The longitudinal expansion and contraction in ferromagnetic rod produces ultrasonic sound waves in the medium surrounding the nickel rod. The frequency range of the waves depend on the mode of vibration of the ferromagnetic material and may vary from few hundred to 3,00,000 Hz. Since, the rod vibrates longitudinally; the frequency of the fundamental mode of vibration can be used as



$$f = \frac{m}{2l} \sqrt{\frac{Y}{\rho}}$$

where Y is the Young's modulus of the material and ρ its density and m is mode of vibration. m=1 fundamental frequency, m=2 first overtone. Similarly the frequency of LC oscillatory circuit is

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$$

3.16 Magnetoresistance

It was discovered by William Thomson in 1856

The change in the electrical resistance of certain materials under the application of external magnetic field is known as magnetoresistance. The materials which exhibit this nature are known as the magnetoresistors. The resistance of the magnetoresistors are directly proportional to the magnetic field.

Working Principle: In the absence of the magnetic field, the charge carriers of the magnetoresistor move in the straight path. But when the field is applied, the electrons will experience a force, which will increase the path length and the collisions of electrons. This will result in increase in the resistance of the material. These magnetoresistors are classified into three categories.

1. Giant magnetoresistors (GMR): In these magnetoresistors, the resistance will become small when their ferromagnetic layers are parallel to each other and vice versa.

2. Extraordinary magnetoresistors: In these magnetoresistors, the resistance of the material is high in the absence of magnetic field and low in the presence of a field.

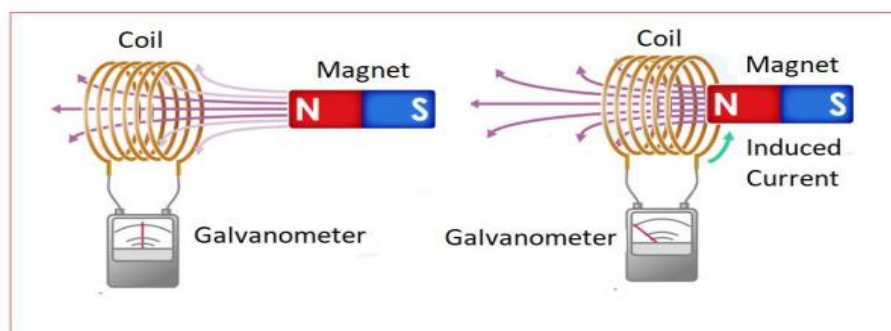
3. Tunnel magnetoresistors: In these magnetoresistors, the heavy current flows if the magnetisation of the electrodes is parallel to each other. The antiparallel arrangement increases the resistance between the layer.

3.17 Magnetic Field Sensors

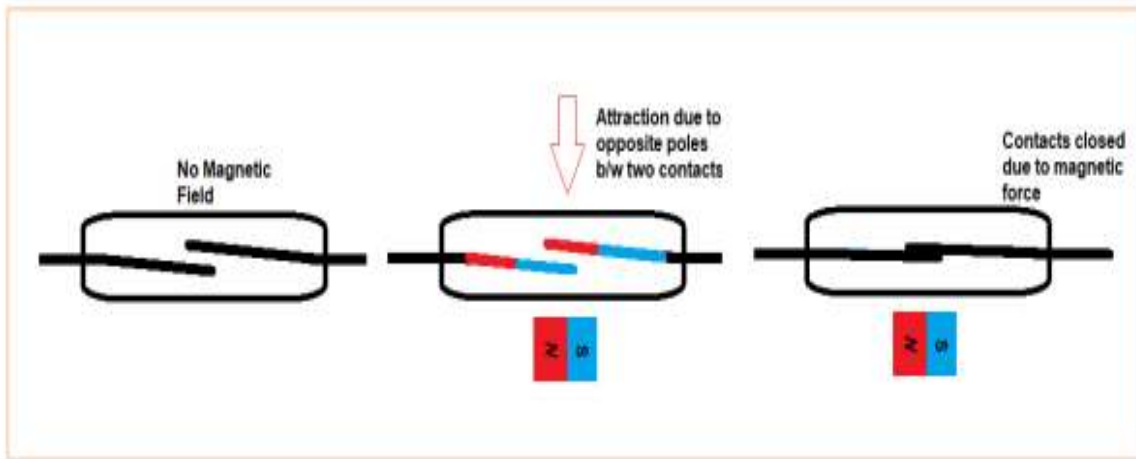
Magnetic field sensors are the devices that detect and measure magnetic fields around permanent magnets, current conductors, and electrical devices. These devices are very accurate, space efficient and operate with little power. These magnetic sensors work by converting the variations of a magnetic field and its magnitude into electric signals.

Typical magnetic sensors and their application

a) Coils: The coil is used to detect the variations in a magnetic field. When a magnet is moved towards a coil, the magnetic flux density in the coil will increase. This increase in magnetic flux density in the coil results in the induced electromotive force and induced current. The measurement of the induced electromotive force and current will allow you to detect the ratio of change in magnetic flux density and its direction. These coils will become highly sensitive magnetic sensors when combined with other coils or magnetic materials. The typical examples for the coil sensors are search coils, rotation angle sensors (resolvers), fluxgate sensors.



b) **Reed switches:** The device consist of a glass tube filled with inert gas to prevent the activation (deterioration) of the contacts.



These glass tube consist two reeds (contacts), made of nickel or other magnetic material and come from the right and left ends of the tube. The role of the inert gas is to prevent the deterioration of the contacts. In the absence of the field, the reed switch is open. But under the application of magnetic field, the magnetic material is magnetized and the contacts are attracted to each other closing the circuit (conduction state).

Unlike semiconductor sensors such as MR sensor elements or Hall elements, the reed switch operates without a power supply and is therefore often used in automobiles or other locations where power is difficult to supply.

c) **MR (Magnetoresistance) sensor elements:** These sensors works on the principle of magnetoresistance effect. There are a number of MR sensor types using different operating principles.

d) **Anisotropic Magneto-Resistance (AMR) sensor elements:** It is discovered by William Thomson in 1956. Anisotropic Magneto-Resistance effect is exhibited by the ferromagnetic material when placed in an external magnetic field. When the direction of magnetization in a ferromagnetic material is parallel to the current, the electron orbital becomes perpendicular to the current. This will increases the spin-dependent scattering of electrons and the resistance of the material will increases.

Similarly when the direction of magnetization is perpendicular to the current, the electron orbital becomes horizontal to the current. This will reducing the spin-dependent scattering and the resistance of the material will decreases. The rate of change in resistance caused by the state of the magnetic field is called magnetoresistive ratio (MR ratio). The MR ratio for an AMR sensor

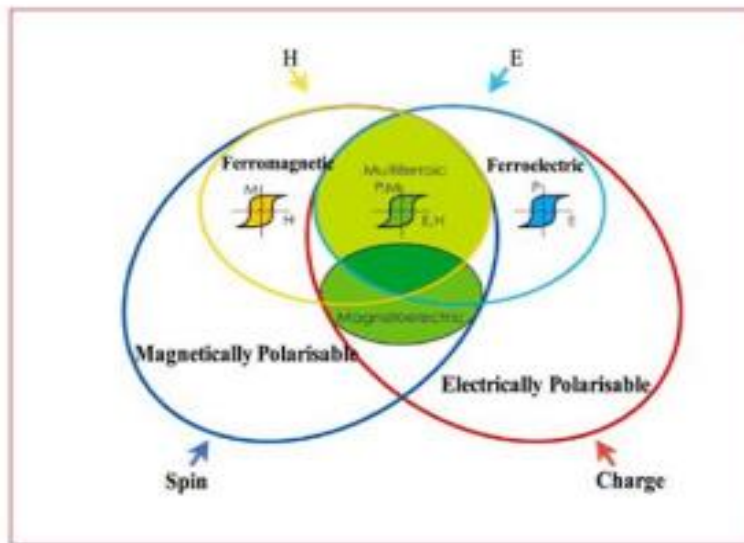
element is about 5%. The AMR sensor element is often used in magnetic switches and rotation sensors because of its simple structure.

3.18 Multiferroics (MFs)

There three fundamental ferroic behaviours in nature, namely

- Ferromagnetism
- Ferroelectricity
- Ferroelasticity

Multiferroics are those materials which will exhibit more than one ferroic behaviour simultaneously. However the multiferroics which will exhibit ferromagnetism and ferroelectricity are more popular and have vast applications. In these materials the intensity of magnetization can be controlled by the electric field intensity and polarization can be controlled by the magnetic field. These multiferroics are classified in two categories i.e. type-I multiferroics and type-II multiferroics



Type-I multiferroics: These materials show a linear magnetoelectric response, as well as changes in dielectric susceptibility at the magnetic phase transition. In these materials the ferroelectricity and ferromagnetism will occur at different temperatures. The ferroelectricity (which requires the structural distortion) occurs at high temperature but the magnetic ordering (usually anti ferromagnetic) occurs at low temperature.

Ex: BeFO₃ (Beryllium Fluorate)

Type-II multiferroics: These are the materials in which the magnetic ordering breaks the inversion symmetry and directly causes the ferroelectricity. In these materials the ferroelectricity and ferromagnetism will occur at same temperatures.

Ex: TbMn₂O₅ (Terbium Manganes Pentoxolane)

Applications : Multiferroics have potential for applications as actuators, switches, magnetic field sensors and new types of electronic memory devices.

ENERGY MATERIALS

Introduction

Energy materials are the class of materials, which will store the energy, convert the energy in one form to another form. The energy materials with dual conversion capacity i.e. electrical energy to chemical energy and chemical energy to electrical energy have striking applications in electronics and automobile industry.

3.19 Electrolytes

Liquid Electrolytes: These class of electrolytes conducts electricity when dissolved in water. These electrolytes, when dissolved in a solvent, dissociate into cations (positive ions) and the anions (negative ions). The conductance of electrolytes is due to ions and termed as ionic conductivity. Conductivity (or specific conductance) of an electrolyte solution is a measure of its ability to conduct electricity.

$$\text{Resistance, } R = \rho \frac{l}{A}$$

L is the length between the electrodes, A is the cross-sectional area of the sample and ρ is the specific resistance.

In practice, the conductivity cell is calibrated by using solutions of known specific resistance, ρ^* , and the cell-constant, defined as the ratio of l and A is derived.

$$\text{Resistance, } R = \rho C, \text{ where } C = \frac{l}{A}, \text{ cell constant.}$$

Hence we can write the specific conductance, the reciprocal of specific resistance (resistivity) as

$$\kappa = \frac{1}{\rho} = \frac{C}{R}$$

Sometimes the conductance (reciprocal of the resistance) is denoted as $G = 1/R$

Hence, the specific conductance can be written as $\kappa = CG$

The specific conductance of a solution containing one electrolyte depends on the concentration of the electrolyte. The molar conductivity is defined as the ratio between the specific conductance (κ) and concentration of the electrolyte(c).

$$\text{molar conductivity, } \Lambda_m = \frac{\kappa}{c}$$

Factors affecting Electrolytic Conductance

1. Concentration of ions: The conductivity of electrolytes is proportional to concentration of ions but molar conductivity increases with a decrease in the concentration of the solution.

2. Nature of electrolyte: The electrolytes which fully dissolve in solution are termed as strong electrolytes. Materials such as KNO_3 , with a higher degree of separation, will fully dissolve in solvents, will have more number of ions in the solution and hence will have higher conductivity. The electrolytes which partially dissolve in solution are termed as weak electrolytes. Materials

such as CH_3COOH , with a small degree of separation, will partially dissolve in solvents, will have less number of ions in the solution hence will have lower conductivity.

3. Temperature: The solubility of an electrolyte there by the concentration increases with increase in temperature.

Solid Electrolytes

Solid ionic conductors that can be used in electrochemical cells as an electrolyte are called solid electrolytes. Only one ion is mobile in solid ionic conductors. The transference number is defined as the ratio of the electric current derived from the cation to the total electric current. According to Patterson, the solids with a transference number > 0.99 termed as solid electrolytes.

A large transference number can reduce concentration polarization of electrolytes during charge–discharge steps, and thus produce higher power density. Solid electrolytes possess a much higher thermal stability, and this makes the solid-state battery (SSB) one of the best choices for the next generation of batteries. Moreover, inorganic solid electrolytes can work in hostile environments, such as in the temperature range from -50°C to 200°C or even higher. The solid-state batteries based on solid-state electrolytes could also significantly increase the energy density and the power density, which are the critical features for large energy storage systems.

Based on their components, the solid-state electrolytes can be divided into three categories

- Inorganic electrolytes,
- Organic polymer electrolytes
- Composite electrolytes

1. Inorganic electrolytes: Inorganic electrolytes usually exhibit high ion conductivity, in the range of 10^{-4} Scm^{-1} . The inorganic electrolyte is more suitable for rigid battery design, as it possesses better thermal/chemical stabilities, higher mechanical strength and exhibits an obvious conductivity advantage over a wide temperature range. Some of the reported inorganic electrolytes show ionic conductivity that is competitive with or even higher than commercial liquid electrolytes. However, performance of solid state batteries with inorganic electrolytes is limited due to the poor interfacial contact between electrolyte and electrode

2. Organic polymers electrolytes: Organic polymers electrolytes are different from inorganic ceramics in their chemical and mechanical properties, which enforces a different battery designs. Organic polymer electrolytes have several advantages over the inorganic solid electrolyte, such as its high flexibility, the ability to easily form intimate electrode-electrolyte contact and the simple process ability. However, these materials face other issues, such as low Li^+ conductivity ($<10^{-5} \text{ Scm}^{-2}$ at room temperature), large ionic transfer resistance, and oxidation at high voltages.

The polymer electrolyte materials are further classified into

- 1 Gel polymer electrolyte and
- 2 Solvent-free solid polymer electrolyte

Gel polymer electrolyte exhibits weaker mechanical properties but higher ionic conductivity. Its conduction mechanism is similar to the liquid electrolyte.

The Solvent-free solid polymer electrolyte exhibits stronger mechanical properties and could be prepared into free-standing electrolyte membranes without adding other mechanical supports. In these materials, Na-salts are dissolved in the polymer host and these Na ions could transport within the polymer hosts via the segmental motion of the polymer chains

3. Composite electrolytes: A composite material is made from two or more constituent materials which have dissimilar chemical or physical properties. The composite material will exhibit the properties different from the individual elements. However, the individual elements remain separate and distinct, distinguishing composites from mixtures and solid solutions. Ceramic/polymer composite electrolytes may solve the ionic conductivity and interfacial contact issues consistently by combining the advantages of two different types of electrolytes.

Requirements of a good solid state electrolyte:

- (i) High ionic conductivity ($>10^{-4} \text{ Scm}^{-2}$)
- (ii) Good mechanical strength and enough structural defects to prevent lithium dendrite penetration;
- (iii) Low-cost raw material and low cost preparation techniques
- (iv) Low activation energy for lithium-ion diffusion.

Experimental studies showed that the ceramic materials can exhibit higher ionic conductivity than polymers. Besides, the electrochemical stability window of polymeric electrolytes is relatively low compared to that of inorganic electrolytes. Hence the researchers are paying much attention to inorganic electrolytes for application in all solid-state batteries. In a crystalline solid electrolyte, thermal energy drives the migration of lithium ions through cationic vacancies or interstitials.

If q the charge of transfer ions, n is the concentration of electrolytes, and mobility of charge carriers of the crystal, σ is given as

18.5

The above equation is referred as modified Arrhenius equation where m is known as activation energy is pre-factor. The typical value of $m = 1$. The above equation shows that increasing the concentration of electrolytes (n) increase the ionic conductivity in the solid. Hence for a specific battery operating temperature, concentration of electrolytes (n) highly related to the mobile defect formation energy (E_f).

Advantages of Solid-state electrolytes over the conventional organic liquid electrolytes

1. Solid-state electrolytes possess higher thermal stability.
2. They will ensures higher safety due to their lower flammability.
3. Durability of Solid-state electrolytes is more.
4. Their battery design is simple.

Disadvantages of Solid-state electrolytes over the conventional organic liquid electrolytes

The room-temperature ionic conductivity in solid electrolytes is lower than that in liquid electrolytes.

3.20 Superionic conductors

Superionic conductors are a class of solid materials which show an unusual high ionic conductivity. Their structure is characterized by strong disorder in the sub-lattice of conducting ions. These solids will have high ionic conductivity and negligible electronic conductivity. The ionic conductivity occurs due to the motion of ions, whereas the electronic conductivity results from the flow of electrons.

Super ionic conductors conductivity (σ) will be usually more than $10^{-2} \Omega^{-1} \text{ cm}^{-1}$.

Few ionic conductors attain high electrical conductivity only above a certain temperature. This is due to results in the generation of anion Frenkel disorder and it appears that both vacancies and interstitials are mobile in the superionic phase.

The Process of ionic conduction: The conduction in the ionic conductors' may due to the presence of crystal defects in solids (Normal Ionic Conductors) or may be due to molten defects (Super ionic conductors).

Normal Ionic Conductors: In normal ionic conductors, the ion transportation is due Frenkel defects are schottky defects. As these defect concentration increases with temperature, the ionic conductivity of these solids increases with temperature.

Super ionic conductors: In the solids with molten defects, all the ions are available for conduction. The number of defect or void sites in these solids is more than the number of ions. This will allow the ions to move freely with low activation energy which will results in high conductivity. These are called superionic conductors (SICs)

The differences between the normal ionic conductors and super ionic conductors

1. Normal Ionic conductors

1. low ionic conductivity and may have appreciable electronic conductivity
2. Conductivity increases with temperature just below melting point.
3. Number of charge carriers are low and conductivity is strongly temperature dependent
4. The activation energy for ion migration is high.
5. The conduction is due to the thermal generated defects and the activation processes involves both energy of defect formation and defect migration

2. Super Ionic conductors

1. High ionic conductivity and negligible electronic conductivity
2. High ionic conductivity even well below the melting point
3. Number of charge carriers are high and conductivity is temperature independent
4. The activation energy for ion migration is low
5. The conduction is due to high carrier concentration and the enthalpy of formation is almost zero

Classification of Fast Ionic Conductors

1. Cationic conductors: In cationic conductors, the high ionic conduction is due to the positively charged mobile ions. The cationic conductors usually contain monovalent cations such as Ag^+ , Na^+ , Li^+ , K^+ or H^+ or divalent cations such as Ca^{2+} , Ba^{2+} , Cd^{2+} , trivalent cations are Nd^{3+} , Eu^{3+} , Er^{3+} , etc. However, it is observed that monovalent ions exhibit high conductivity. The conductivity changes with temperature as the structure of crystal changes with temperature. For example, the Ag^+ conductivity in AgI is low At room temperature because I^- ions adopt either hexagonal or cubic-close-packed structure. When the temperatures is raised above 146°C , AgI transforms into a polymorph in which I^- ions adopt an open body-centered-cubic packed structure containing disordered Ag^+ ions and the conductivity of AgI increases sharply to about 1 S cm^{-1} .

2. Anionic conductors: In the anionic conductors, the high ionic conduction is due to the negatively charged mobile ions. Anionic conductors are classified in to oxide (O^{2-}) and fluoride (F^-) groups. Conductors, such as stabilized zirconias and PbF_2 , require high temperatures for fast ion conductivity. These conductors usually have the fluorite (CaF_2) type structure in which anions (O^{2-} or F^-) are arranged at corners of a cube and cations occupy body centers of alternative cubes.

3. Amorphous conductors: In addition to crystalline inorganic solids, superionic conductivity can also be found in amorphous inorganic materials. Glassy electrolytes based on sulfides and halides have much higher conductivity than oxides at ambient temperatures. Many Li^+ and Ag^+ conducting glasses are known prepared and had vast applications. Even though oxide-based glasses have low conductivity, they are more stable than sulfides or halides at high temperatures

4. Polymer-based conductors: Superionic conductivity is also known in polymer based solids. The polymer electrolytes are classified in to two types.

1. Polymer-salt complexes

2. Polyelectrolytes.

- Polymer-salt complexes are prepared by dissolving a salt such as LiClO_4 in a polymer such as poly ethylene oxide. Both cations and anions in polymer-salt electrolytes can be mobile.
- In polyelectrolytes, the polymer backbone contains covalently attached charged (positive or negative) groups and the charge-balancing counter ions are able to make long-range migration. The best-known polyelectrolyte is Nafion perfluorocarbon sulfonic acid polymer, which is a proton conductor under wet conditions.

Applications of Superionic conductors

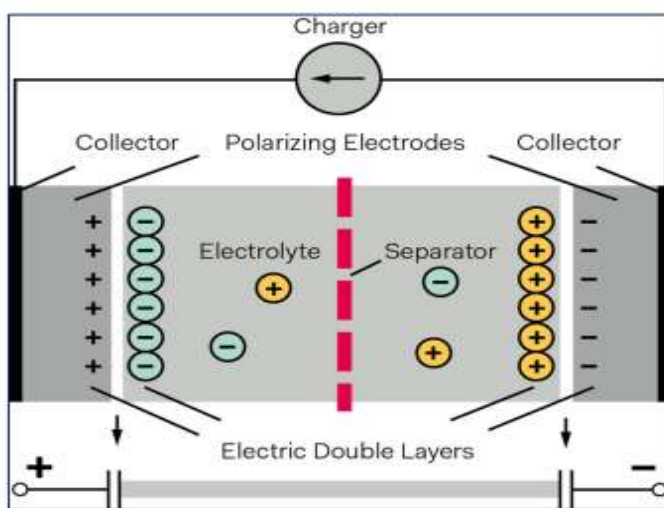
- Energy storage and generation (electrolyzers, batteries, and fuel cells), Membranes in ion selective electrodes
- For the development of all-solid-state electrochemical devices, which have many advantages over those based on liquid electrolytes including ease of miniaturization and high-temperature stability.

- Gas Sensors: Oxygen gas sensors based on oxide conductors are being widely used to monitor automobile exhaust gases. Oxide conductors are also being studied for the construction of solid oxide fuel cells (SOFC).
- Nafion and related sulfonated polymers are under investigation as proton conductors in polymer electrolyte membrane (PEM) fuel cells
- Electrochromic display capacitors,
- Analog memory devices,
- Miniature cells,
- Power storage devices

3.21 Materials and electrolytes for super capacitors

The super capacitors (SCs) are the class of capacitor with capacitance value much higher than other capacitors, but with lower voltage limits. These super capacitors bridges the gap between electrolytic capacitors and rechargeable batteries. The various electrode material used for making the supercapacitors are Graphene, polymer, metal oxides, carbon etc. Different electrolytes like Li, Na, Na₂SO₄, sulphuric acid and phosphoric acid used in super capacitors.

Construction and working of a supercapacitor



A supercapacitor cell consists of two electrodes, a separator, and an electrolyte. The collectors are made with the metals of high conductivity. A membrane is kept between the two electrodes which allow only the ionic conductivity, but stops electrical conductivity. An active material, made up of metal oxides, carbon and graphite is used as an electrolyte. The system is impregnated with an electrolyte. The geometrical size of the two carbon sheets and of the separators

is designed to have huge surface area. Due to its structure, the highly porous carbon can store more energy than any other electrolytic capacitor. Under the application of voltage, the positive plate attracts negative ions from the electrolyte, and negative plate attracts positive ions. This will form the ion layers to the either side of the plate (termed as double layer formation), resulting in the ions being stored near the surface of the carbon. This mechanism gives supercapacitors the ability to store and restore high energy in a very short time period.

The advantages of supercapacitors

- Their storage capacity is about 10 to 100 times more per unit volume compared to electrolytic capacitors.
- They also have the advantage of fast charging and discharging
- Higher charging and discharging cycles compared to rechargeable batteries.

Applications of Supercapacitors

- Used in applications requiring many rapid charge/discharge cycles, rather than long-term compact energy storage.
- In automobiles, buses, trains, cranes and elevators, where they are used for regenerative braking → Short-term energy storage
- burst-mode power delivery
- As power backup for static random-access memory (SRAM).

Unlike ordinary capacitors, supercapacitors do not use the conventional solid dielectric. The two storage principles by which the electrical energy is stored in the supercapacitors are

- Electrostatic double-layer capacitance
- Electrochemical pseudocapacitance

The distribution of the two types of capacitance depends on the material and structure of the electrodes.

1. Electrostatic double-layer capacitors (EDLCs): Electrostatic double-layer capacitors (EDLCs) use carbon electrodes or derivatives with much higher electrostatic double-layer capacitance than electrochemical pseudocapacitance, achieving separation of charge in a Helmholtz double layer at the interface between the surface of a conductive electrode and an electrolyte. The separation of charge is of the order of a few ångströms (0.3–0.8 nm), much smaller than in a conventional capacitor.

2. Electrochemical pseudocapacitors: Electrochemical pseudocapacitors use metal oxide or conducting polymer electrodes with a high amount of electrochemical pseudocapacitance additional to the double-layer capacitance. Pseudocapacitance is achieved by Faradaic electron charge-transfer with redox reactions, intercalation or electrosorption. Hybrid capacitors, such as the lithium-ion capacitor, use electrodes with differing characteristics: one exhibiting mostly electrostatic capacitance and the other mostly electrochemical capacitance. The electrolyte forms an ionic conductive connection between the two electrodes which distinguishes them from conventional electrolytic capacitors where a dielectric layer always exists, and the so-called electrolyte, e.g., MnO₂ or conducting polymer, is in fact part of the second electrode (the cathode, or more correctly the positive electrode). Supercapacitors are polarized by design with asymmetric electrodes, or, for symmetric electrodes, by a potential applied during manufacturing.

3.22 Rechargeable ion batteries

A rechargeable battery is a type of electrical battery which can be charged, discharged into a load, and recharged many times. The three most popular rechargeable batteries are

- Nickel-Cadmium (Ni-Cd)
- Nickel Metal-Hydride (Ni-MH)
- Lithium-Ion (Li-Ion)

Energy Density of a battery

1. The gravimetric energy density: This is the total energy of a battery in comparison to its weight. Units Watt-hours/kilogram (W-hr/kg).

2. The volumetric energy density: This is the total energy of a battery in comparison to its volume. Units: Watt-hours/liter (W-hr/l). The Amp-hour capacity: It is defined as the amount of current that a battery can deliver for 1 hour before the battery voltage reaches the end-of-life point.

Types of lithium-ion batteries

1. Lithium Cobalt Oxide batteries: Lithium cobalt oxide batteries are made up of lithium carbonate and cobalt. They have a cobalt oxide cathode and a graphite carbon anode. The lithium ions migrate from the anode to the cathode during discharge, with the flow reversing when the battery is charged. These batteries are used in cell phones, laptops, and electronic cameras because of their high specific energy. These batteries has a few disadvantages, including short battery life and low specific power.

2. Lithium Manganese Oxide Batteries: These batteries have high-temperature stability compared to the others which makes them safer than other lithium-ion battery types. Due to this property, they are used in medical equipment and devices, power tools, electric motorcycles, and other applications.

3. Lithium Iron Phosphate (LFP) Batteries: Phosphate is used as a cathode in lithium ion phosphate batteries. They have low resistance qualities, which improve their thermal stability and safety. They have high durability and a long lifecycle. Fully charged batteries can be kept with no effect on the battery's overall charge life. Due to the lengthy battery life, Li-phosphate batteries are cost-effective. However, due to the lower voltage of the lithium-phosphate battery, they have less energy in comparison with other lithium batteries. As a result, these batteries are frequently used in electric bikes and other applications requiring a long lifecycle and high levels of safety. These batteries are also commonly used in electric vehicles. These batteries are frequently used in electric vehicles and other applications requiring a long lifecycle and high levels of safety.

4. Lithium Nickel Manganese Cobalt Oxide (NMC) batteries: Lithium Nickel manganese Cobalt Oxide batteries are constructed with a variety of materials. A cathode constructed of a mix 60% nickel, 20% manganese, and 20% cobalt. These batteries are cost effective as nickel is cheaper. These batteries can have a high specific energy density or a high specific power (but

can't have both features). Because of their low self-heating rate, they are most commonly seen in power tools, electric vehicles.

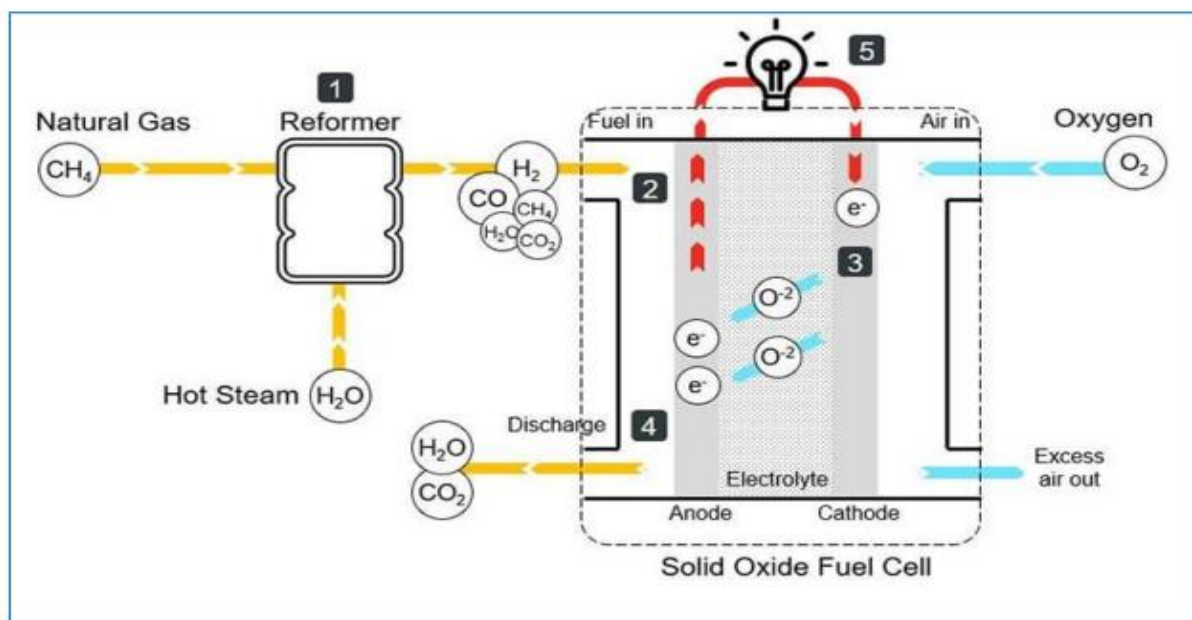
5. Lithium Nickel Cobalt Aluminum Oxide batteries: These batteries offer a high-energy density with a long lifespan. However they are less safe than other lithium-ion battery types and are rather expensive. Hence, while these batteries are used in electrical vehicles, they must be accompanied by monitoring systems to ensure driver safety. These batteries have little applications in consumer electronics, but have great potential in the automobile industry. Given the widespread use of NCA batteries in electric vehicles, demand for these batteries may increase as the number of electric vehicles grows.

6. Lithium Titanate Batteries: These batteries are made using the nanotechnology and have an extremely rapid recharge time. However, their lower energy density than other lithium-ion battery types (but more than non-lithium-ion batteries) is the disadvantage while using them in power automobiles. Li-titanate batteries are being used in electric vehicles and bicycles. The researchers are working to increase their power density to extend their applications in public transportation.

3.23 Solid Oxide Fuel Cells (SOFC)

Fuel cells convert the chemical energy stored in the fuel to electric and thermal energy without the need for combustion. This makes them different from engines and power plants, which may have the same applications but rely on combustion, reducing the efficiency of the energy conversion.

Working Principle of a Solid Oxide Fuel Cell:



Natural gas goes through a steam-reforming process. This chemical reaction produces hydrogen (H_2), carbon monoxide (CO), carbon dioxide (CO_2) and steam (H_2O). This mix of elements

enters the fuel cell at the anode side while the air enters the fuel cell at the cathode side. The oxygen in the air combines with free electrons to form oxide ions at the cathode. These oxide ions along with the free electrons travel from the cathode to the anode through the electrolyte. At the anode, oxide ions react with hydrogen forming water (steam) and with carbon monoxide (CO) forming carbon dioxide (CO₂). This reaction will release free electrons, which travel to cathode through the external electrical circuit, producing electricity.

Applications of Solid Oxide Fuel Cells.

1. High efficiency delivers financial benefits
2. Electrical efficiency of solid oxide fuel cells is close to 60% which is much higher than the efficiencies of coal power plants.
3. As the fuel cells are located locally, the distribution losses will be minimum, unlike the central power plants.
4. This high efficiency delivers financial benefits and minimizes the environmental footprint, since
5. The solid oxide fuel cells are environmental friendly as they use natural gas as fuel in comparison to traditional power plants using coal as fuel.
6. Solid oxide fuel cells also don't emit sulphur oxides and particulate matter.
7. Modular design brings scalability: The individual fuel cells are bundled together to form a stack. These stacks are then combined with other equipment to form modules. These individual power generation modules can be paralleled to form the fuel cell power system. You can add more fuel cell modules to the overall system as you need. This provides financial flexibility for the user to align the power generation investments with business needs.

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Short questions

1. Define the terms: dielectric constant, electric displacement, polarizability and electric susceptibility.
2. Deduce Clausius-Mosotti relation.
3. Explain ferroelectricity.
4. Explain piezoelectricity.
5. Define the terms: magnetic field strength, magnetic flux density, magnetization and magnetic susceptibility.
6. Derive relation between χ_m and μ_r .
7. Why diamagnetic materials have negative susceptibility?
8. What are soft and hard magnetic materials?
9. What are the applications of ferrites?
10. Write about the hysteresis curve?
11. What is electrolyte?
12. How the electrolytes are classified?
13. Mention the advantages of Solid electrolytes over the liquid electrolytes
14. Mention the requirements of a good electrolyte
15. Differentiate between the normal ionic conductors and super ionic conductors
16. What are supercapacitors? Write few applications of supercapacitors

Essay type questions

1. Define current and current density and derive continuity equation.
2. What are Maxwell's equations? Write Maxwell's equations and explain their significance.
3. What is internal field of a dielectric and how it is calculated for a dielectric?
4. Classify and explain the properties of different types of magnetic materials.
5. What is domain? Explain ferromagnetic hysteresis on the basis of domains.
6. What are anti-ferro and ferri magnetic materials? Explain their properties.
7. Write a detail notes on solid and liquid electrolytes
8. Write a detail note on superionic conductors.
9. Write a detail notes on rechargeable batteries
10. What are Solid Oxide Fuel Cells? With neat diagram, explain the working principle of Solid Oxide Fuel Cells

Multiple answer questions

1. The expression for local field (E_{local}) of a dielectric material is given by []
 (A) $E_i = P + \left(\frac{E}{3\epsilon_0}\right)$ (B) $E_i = E + \left(\frac{E}{3\epsilon_0}\right)$
 (C) $E_i = P + \left(\frac{3\epsilon_0}{E}\right)$ (D) $E_i = E + \left(\frac{P}{3\epsilon_0}\right)$
2. Orientational polarizability []
 (A) increases with increase of temperature (B) decreases with increase of temperature

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- (C) is independent of temperature (D) becomes zero with increase or decrease of temp.
3. The Classius-Mosotti equation is []
- (A) $\frac{(\epsilon_r - 1)}{(\epsilon_r + 2)} = \frac{3\epsilon_0}{N\alpha}$ (B) $\frac{(\epsilon_r - 1)}{(\epsilon_r + 2)} = \frac{N\alpha}{3\epsilon_0}$ (C) $\frac{(\epsilon_r + 2)}{(\epsilon_r - 1)} = \frac{N\alpha}{3\epsilon_0}$
- (D) $\frac{(\epsilon_r + 2)}{(\epsilon_r - 1)} = \frac{3\epsilon_0}{N\alpha}$
4. In the absence of external electric field, the electric dipoles in a dipole substance are []
- (A) Parallel (B) Anti Parallel
- (C) Randomly Orented (D) Perpendicular to each other
5. Dielectrics are []
- (A) Solids (B) Metals (C) Semiconductors (D) Insulators
6. The dipole moment per uint volume is []
- (A) Polarization (B) Electric flux density (C) polarization Vector (D) polarizability
7. Choose the correct relation for ionic polarization []
- (A) $\alpha_i = \frac{e^2}{\omega^2 \epsilon_0} \left(\frac{1}{M} + \frac{1}{m} \right)$ (B) $\alpha_i = \frac{e^2}{\omega^2 \epsilon_0} \left(\frac{1}{M} - \frac{1}{m} \right)$
- (C) $\alpha_i = \frac{e^2}{\omega^2 \epsilon_0} (M + m)$ (D) $\alpha_i = \frac{e^2}{\omega^2 \epsilon_0} (M - m)$
8. Material which lack permanent dipoles are called []
- (A) Ferrimagnetic (B) Antiferromagnetic (C) Diamagnetic (D) Paramagnetic
9. The value of permitivity of free space is equal to..... []
- (A) $4\pi \times 10^{-7} \text{Hm}^{-1}$ (B) $8.85 \times 10^{-12} \text{Fm}^{-1}$ (C) $1.68 \times 10^{-19} \text{C}$ (D) $2.54 \times 10^{-23} \text{J}$
10. Choose the correct Relation []
- (A) $E = \epsilon_0 (\epsilon_r - 1) P$ (B) $D = \epsilon_0 (\epsilon_r - 1) E$ (C) $P = \epsilon_0 (\epsilon_r - 1) E$ (D) $\epsilon_r = (\chi - 1)$
11. Hystersisi in polarization-electric field relation is exhibited by []
- (A) Piezo electric materials (B) Ferro electric materials
- (C) Pyro elec tric materials (D) Electro optic materials
12. The polarizations which are dependent of temperature are []
- (A) Orentation and Space Charge (B) Electronic and Ionic

Applied Physics

I -B.Tech

- (C) Orientation and Ionic (D) Electronic and Space Charge
13. In the case of pyroelectric materials changes with temperature. []
(A) Current (B) Dielectric constant (C) Polarization (D) Optical property
14. The dielectric constant of solids []
(A) $\epsilon_r > 1$ (B) $\epsilon_r < 1$ (C) $\epsilon_r = 1$ (D) $\epsilon_r = 0$
15. Using inverse Piezo-electric effect, can be produced. []
(A) Voltage (B) Electric field (C) Current (D) Stress
16. Polarization per unit applied electric field is called []
(A) electric susceptibility (B) magnetic susceptibility
(C) electric polarizability (D) dielectric constant
17. Equal number of opposite spins with different magnitudes present in..... materials []
(A) Ferrimagnetic (B) Antiferromagnetic (C) Diamagnetic (D) Paramagnetic
18. Magnetization per unit magnetic field intensity is called []
(A) magnetic induction (B) magnetic susceptibility
(C) magnetic permeability (D) magnetic flux density
19. Among the following materials, which exhibits ferro-magnetic property? []
(A) Al (B) Gold (C) Nickel (D) Silver
20. The MKS unit of..... is weber per square metre. []
(A) Hm^{-1} (B) Tesla (C) $\text{kg s}^{-2} \text{A}^{-1}$
21. Diamagnetic susceptibility is []
(A) Large, negative (B) Small, positive (C) Small, negative (D) Large, positive
22. The unit of magnetic field intensity is []
(A) Am^{-1} (B) Hm^{-1} (C) Wb^{-2} (D) No units
23. The nuclear magnetic momentum is expressed in the unit of nuclear magneton its value is..... []
(A) $5.05 \times 10^{-29} \text{A-m}^2$ (B) $5.05 \times 10^{-29} \text{A-m}$ (C) $5.5 \times 10^{-29} \text{A-m}$ (D) $5.05 \times 10^{-29} \text{W-m}^2$
24. Super conductors can be called as []
(A) Perfect Diamagnetic (B) Perfect Ferro magnetic
(C) para magnetic (D) Non magnetic material